Limits:

$$\lim_{x \to 0} \frac{x\cos x + \sin x}{5x}$$

$$\Rightarrow \frac{0(\cos(0)) + \sin(0)}{5(0)} = \frac{0}{0} \quad I.F$$

L'Hôpital Rule:

$$(x\cos x + \sin x)' = x(-\sin x) + 2\cos x$$

$$(5x)' = 5$$

$$\Rightarrow \frac{x(-\sin x) + 2\cos x}{5} = \frac{2}{5}$$

limits:

$$\lim_{x \to \frac{\pi}{3}} \frac{\sin(3x)}{\cos(3x) + 1}$$

$$\Rightarrow \frac{\sin\left(\frac{3\pi}{3}\right)}{\cos\left(\frac{3\pi}{3}\right) + 1} = \frac{\sin\pi}{\cos\pi + 1} = \frac{0}{-1+1} = \frac{0}{0}$$

L'Hôpital Rule:

$$(\sin(3x))' = 3\cos(3x)$$

$$(\cos(3x) + 1)' = 3(-\sin(3x)$$

$$\Rightarrow \frac{3\cos(3x)}{-3\sin(3x)} = \frac{3\cos(\pi)}{-3\sin(\pi)} = \frac{0}{3} = 0$$

$$\boxed{\lim_{x \to \frac{\pi}{3}} \frac{\sin(3x)}{\cos(3x)+1} = 0}$$